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CANNY EDGE DETECTION

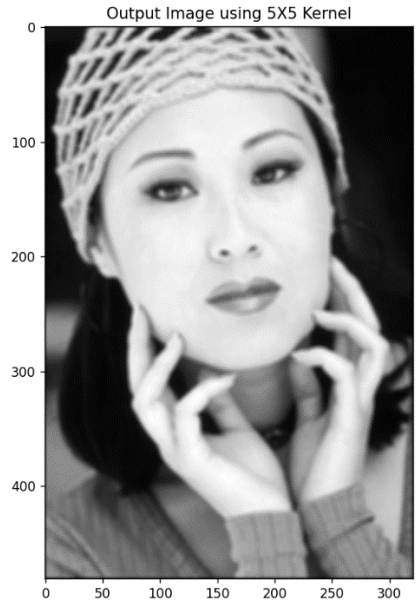
1. **Noise Reduction :**First we have to store the image as a matrix and apply gaussian blur to the image. For edge detection, we have to calculate derivates. Derivatives are also effected by noise, hence it's advisable to smooth the image first before taking the derivative. Then we can use the convolution using the mask to detect the edges.

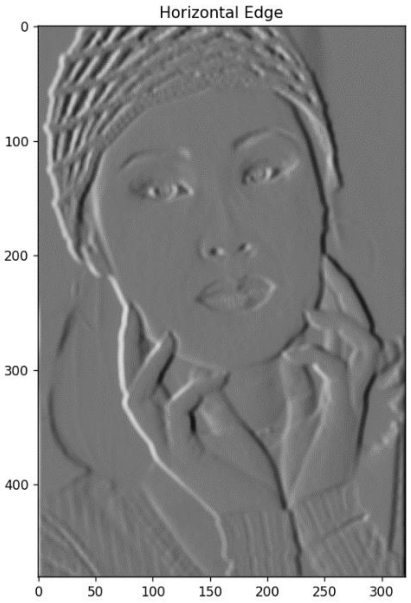
2. **Gradient Calculation :** The Gradient calculation step detects edge intensity and direction using gradient magnitude and direction. When the image is smoothed, the derivatives Ix and Iy.

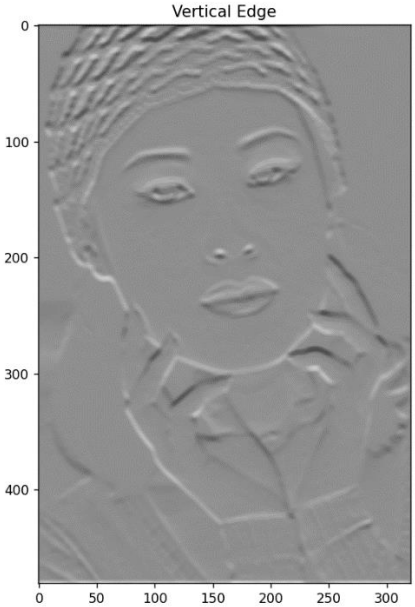
3. **Non max suppression:** This step is implemented to remove redundant edges identified by Sobel Edge Detection. We want to make sure no adjacent pixels are representing the same edge and in order to do that, we compare the gradient magnitude between one pixel and side pixels along the same direction and select the pixel whose magnitude is the largest.

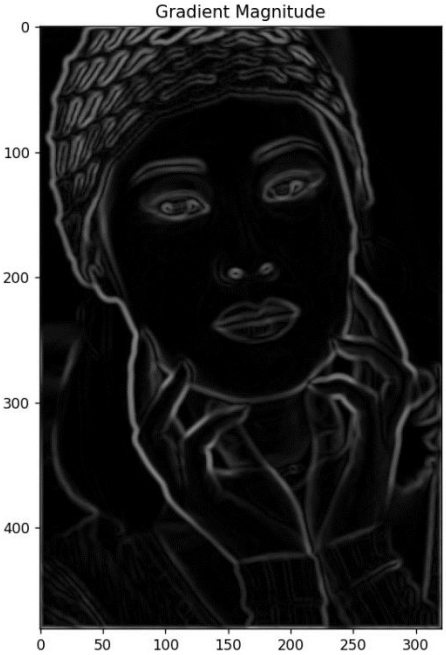
4. **Thresholding:** The main idea of thresholding is to take all the edges and to separate them as weak (here 50) or strong (white 255).

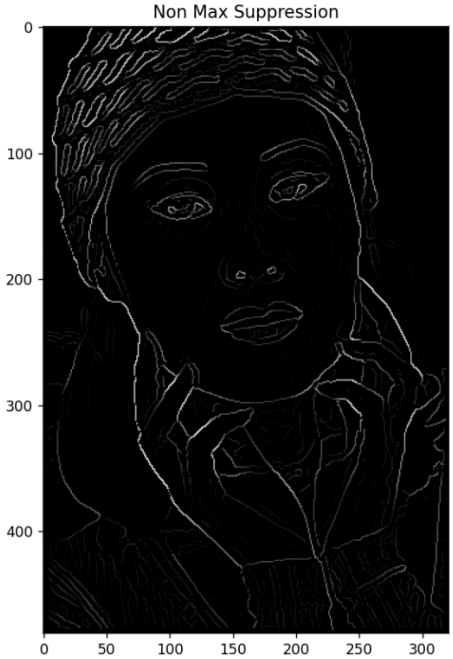
5. **Hysteresis:** We are implementing this to find out whether any given pixels neighbors has value equal to 255, if yes then change the value of the pixel to 255, otherwise discard the pixel by setting the value to 0**.**

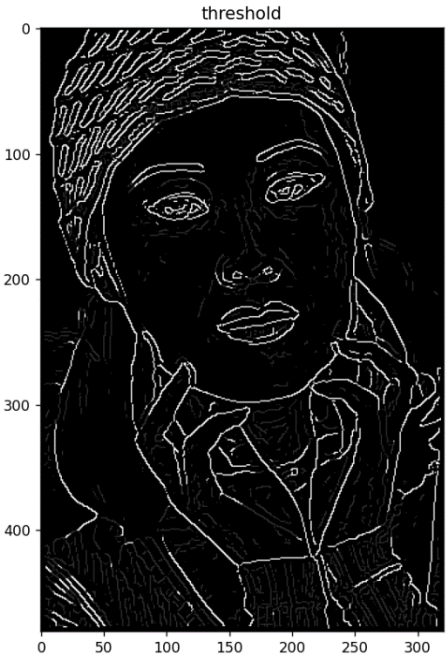

Output Image using 5X5 Kernel
0
100
200
300
400
0 50 100 150 200 250 300

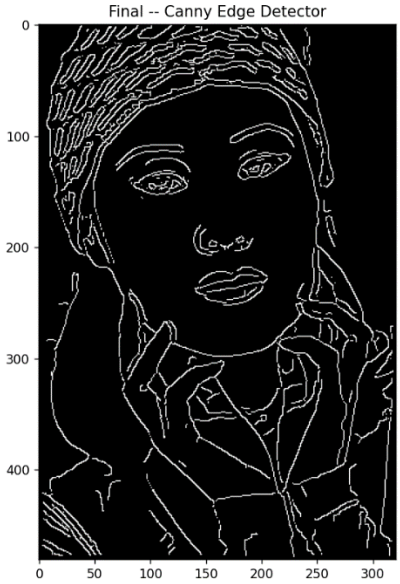

Horizontal Edge
0
100
200
300
400
0 50 100 150 200 250 300


Vertical Edge
0
100
200
300
400
0 50 100 150 200 250 300


Gradient Magnitude
0
100
200
300
400
0 50 100 150 200 250 300


Non Max Suppression
0
100
200
300
400
0 50 100 150 200 250 300


threshold
0
100
200
300
400
0 50 100 150 200 250 300


Final -- Canny Edge Detector
0
100
200
300
400
0 50 100 150 200 250 300

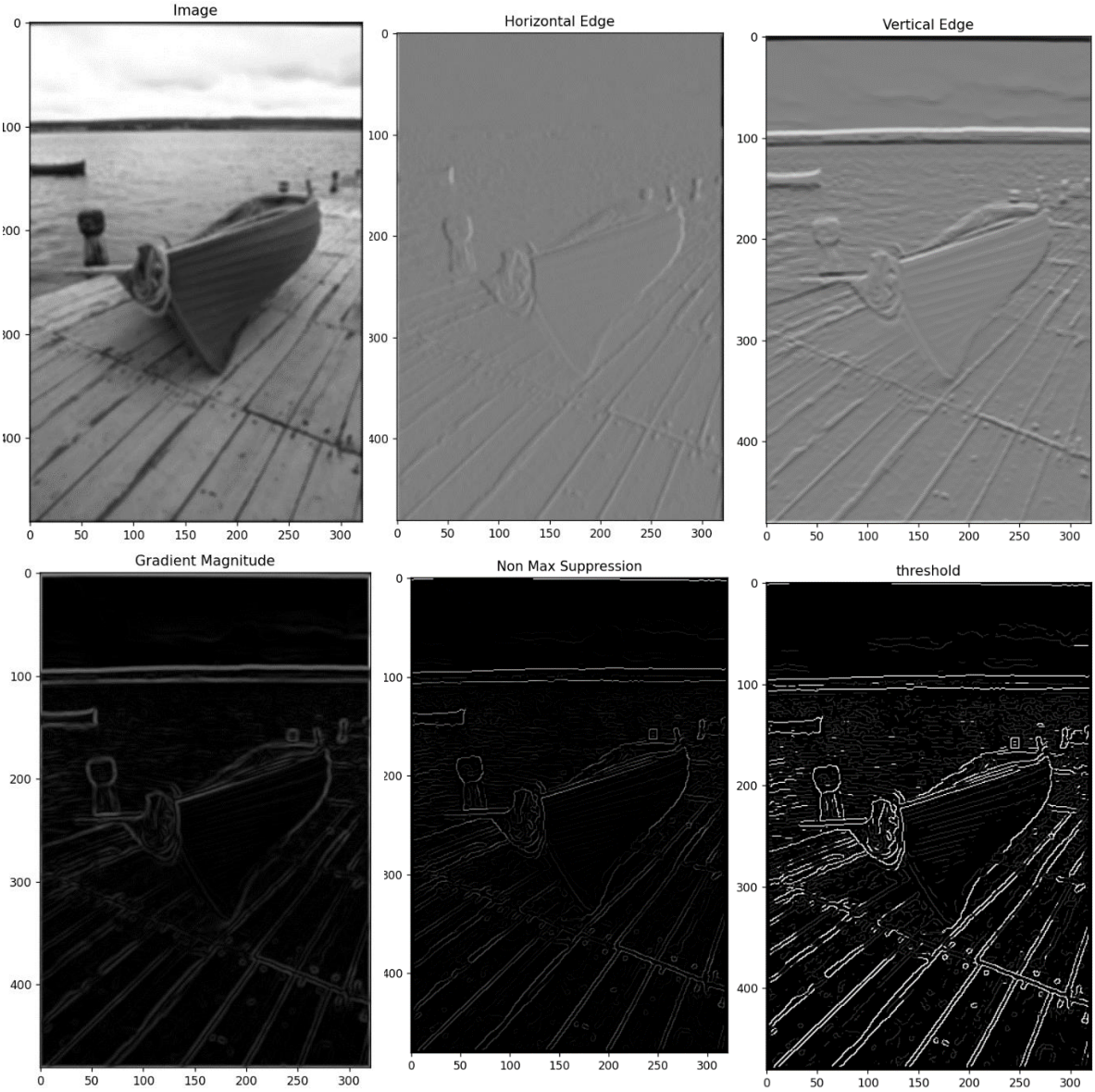
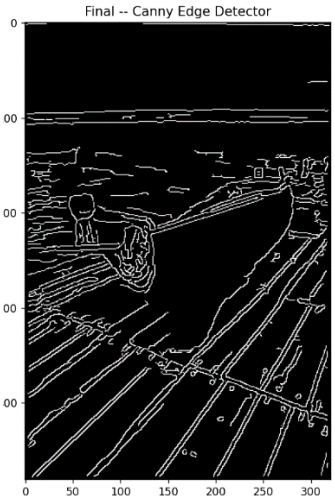
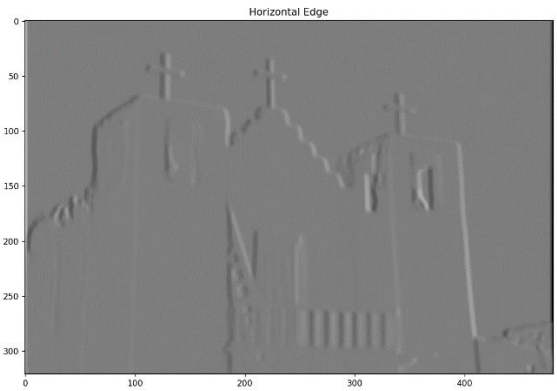
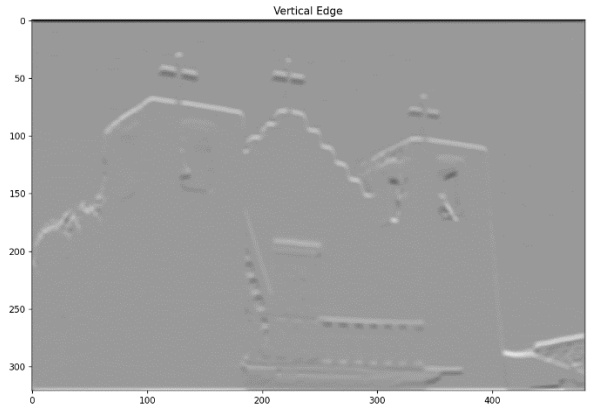


Image
Horizontal Edge
Vertical Edge
Gradient Magnitude
Non Max Suppression
threshold


Final -- Canny Edge Detector

Image

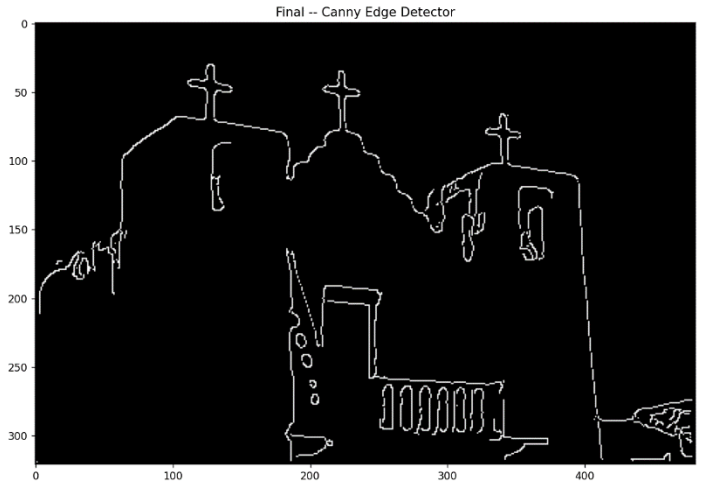

Horizontal Edge


Vertical Edge

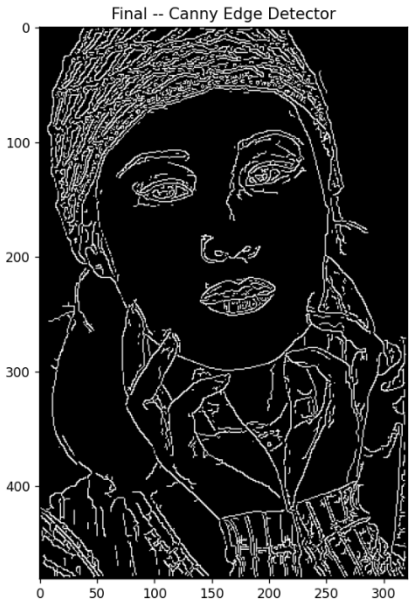
Gradient Magnitude

Non Max Suppression

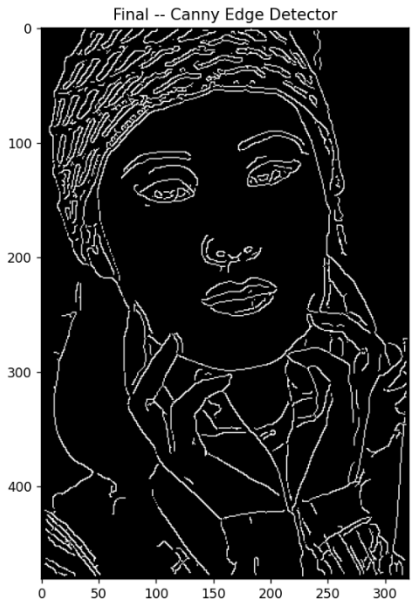
threshold


Final -- Canny Edge Detector

Sigma = 0.5



sigma = 1.5



Sigma = 2.3

- **Best edge detection at sigma value 2.3**